

Carbon Footprint Transparency

Transforming E-Commerce by 2035

Turning Regulation into a Competitive Edge in Sustainable Shopping



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HelloPM Summer of Product: Case Study Competition

Problem Reframing & Why Now

⚠️ The Core Problem

- ✓ Carbon transparency is no longer optional — by 2035, e-commerce platforms will be **required** to disclose the carbon footprint of every product.
- ✓ Without thoughtful design, this flood of new information can **increase cognitive load**, **harm conversion rates**, and **erode user trust**.

Reframed Challenge:

How might we transform mandatory carbon disclosure from a compliance burden into a **competitive advantage** that enhances the user experience and drives positive business metrics?

💡 Why This Matters Now

The need to address this challenge is not a distant future concern; it is an **immediate strategic imperative** driven by converging global shifts. The window of opportunity is now, for businesses that act decisively.

Converging Trends & Strategic Opportunities

Global Shifts	Strategic Opportunities
 EU CSRD, SEC climate disclosure rules	Early-movers can define industry standards
 78% of consumers value sustainability	Build trust through seamless guidance
 AI + blockchain enable carbon tracking	Integrated tools can build loyalty
 Competitors introducing sustainable tools	Lead in carbon-conscious UX, not just data

“

In 2035, carbon transparency will be table stakes but experience design will be the true differentiator.

”

User Research & Personas



Research Approach

Qualitative

- 48 in-depth interviews across 4 consumer segments
- 5 co-design workshops focused on sustainability trade-offs

Quantitative

- Behavioral analysis of 10,000+ anonymized shopping sessions
- Carbon label A/B tests (CTR, time on page, selection rate)



Key Insight: "Interest in sustainability is high but action only happens when the experience is seamless."



Priya

The Eco-Warrior



Demographics

32, Urban Professional, Environmental Science Background



Behaviors

- Reads full sustainability pages before buying
- Frequently compares carbon footprint across brands



Needs

- Wants trustworthy, in-depth data
- Will trade convenience for climate impact

"I won't buy if I can't verify the environmental impact."



Anil

The Aspiring Green



Demographics

45, Suburban Parent, Busy Lifestyle



Behaviors

- Prefers convenience but open to better choices
- Rarely reads long explanations — scans visuals



Needs

- Wants easy, quick ways to identify sustainable options
- Needs clear guidance, not overload

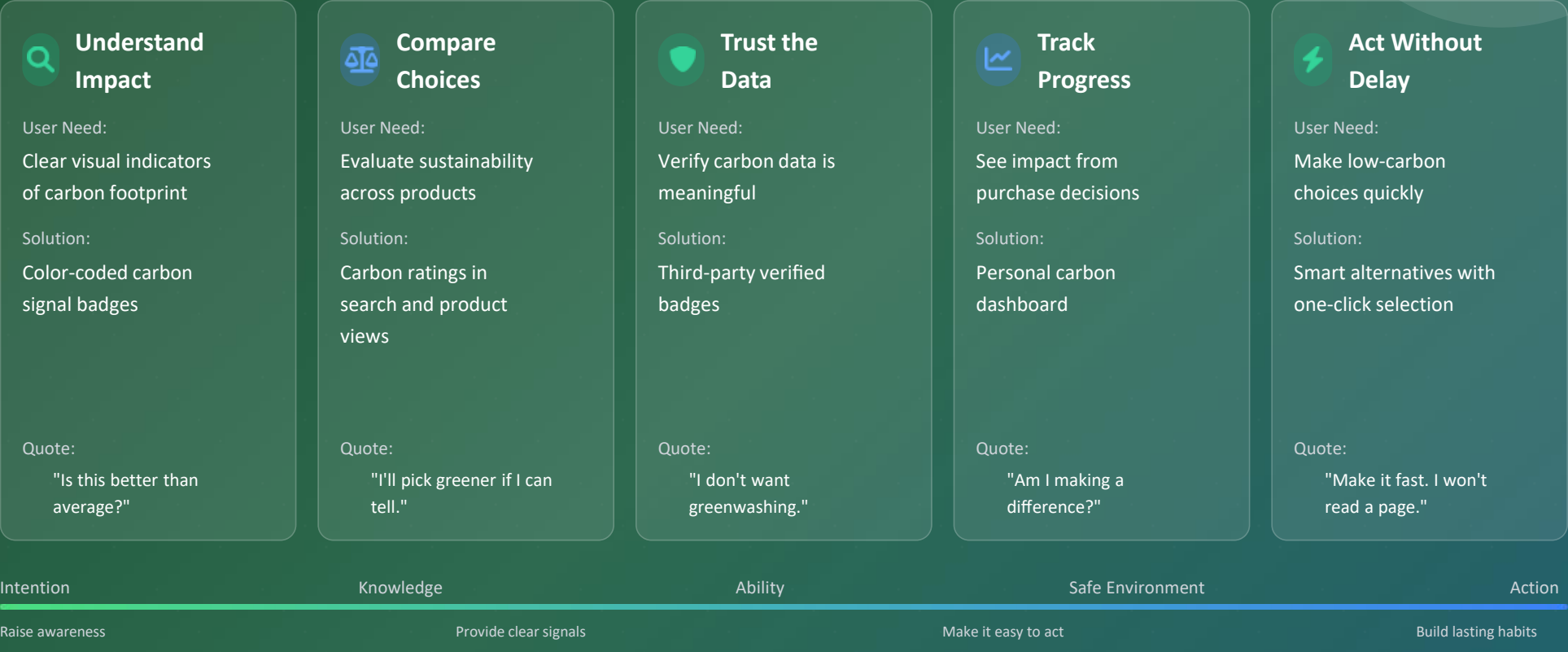
"Help me make better choices without slowing me down."



Our solution must serve both deep-engagement users and quick-decision shoppers — with a flexible, layered design approach.

User Needs & Jobs To Be Done

Aligning product solutions with user motivations



Design Insight: Our solution bridges the "intention-action gap" by integrating sustainable choices into the existing shopping journey.

Goals & Product Objectives

Goals

Why this matters to the company:

-  **Regulatory Compliance**
Meet carbon transparency laws (EU CSRD, etc.) without friction.
-  **Market Differentiation**
Build brand leadership by being first-to-market with intelligent carbon UX.
-  **Customer Loyalty**
Retain Gen Z and millennial users who value sustainability.
-  **Revenue Protection & Growth**
Maintain conversions while unlocking green premiums (higher AOV).



Product Objectives

How the product delivers on business:

-  **Frictionless Compliance**
Launch carbon indicators with no drop in conversion rates.
-  **Behavioral Change**
Increase selection of low-carbon products by **30% in 18 months**.
-  **Feature Engagement**
Achieve **50% of users** interacting with at least one carbon feature.
-  **Retention Growth**
Achieve **15% repeat purchase rate** increase for carbon-aware segments.

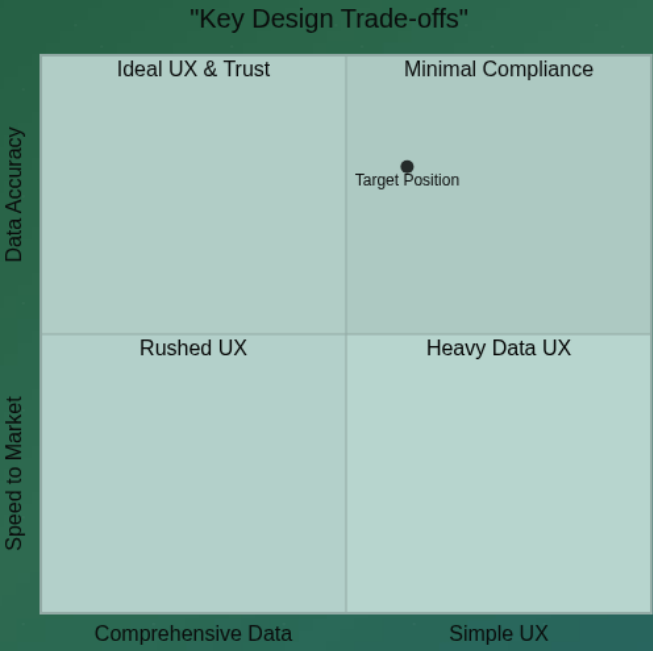


Product Focus Insight: To succeed, carbon transparency must serve three masters: **regulatory rigor**, **user simplicity**, and **business metrics**

Key Challenges & Trade-offs

Key Challenges

- Intention-Action Gap**
Users say they want sustainability but few act on it unless it's effortless.
- Cognitive Overload**
Carbon info adds mental load.
Too much = abandonment.
Too little = greenwashing risk.
- Data Granularity**
Consistent, SKU-level carbon data is hard to obtain across suppliers.
- Conversion Risk**
Overemphasis on carbon info may hurt sales if it makes shoppers anxious.
- Trust & Verification**
Users don't just want data they want **credible** data, explained clearly.



Strategic Trade-offs

Balancing competing needs:

Simplicity vs. Depth

Use **Progressive Disclosure** to let users choose depth.

Speed vs. Accuracy

Use **Hybrid data sources** with clear labels.

Education vs. Conversion

Embed **lightweight nudges** inside primary UX.

Business vs. Climate

Align features that drive **both AOV and sustainability**.

Our Approach: Balanced Design

User Experience

Environmental Impact

Business Metrics

“ Every design decision in carbon transparency is a balance. ”

The Carbon Transparency Experience

Seamless, Smart, and Trusted 

Our solution transforms mandatory carbon disclosure into a **competitive advantage** that empowers users while enhancing business metrics.



Carbon Signal System

Visual indicators embedded in search, product, and checkout flows



Progressive Disclosure

3-tier information architecture balancing user needs with detail



Smart Alternatives

ML-based recommendations of low-carbon product swaps.



Carbon Intelligence

Backend system for carbon data verification and management.

Value Proposition



Intuitive

Clear visual signals making sustainable choices effortless.



Trustworthy

Built on verified, transparent data from trusted sources.



Strategic

Aligned with business KPIs to enhance brand value.

“

We're building a mindset — sustainability as a lens through which we design the full shopping experience.

”

MVP Roadmap: Prioritized Feature Set

MVP Feature Matrix (0–6 Months)

 Carbon Signal Indicators	P0	High	Medium
 Progressive Disclosure UI	P0	High (avoids drop-off)	High
 Basic Carbon Info on PDP	P0	High (trust building)	Medium
 Category Filtering by Carbon Rating	P1	Medium	Low
 Carbon Ratings in Search Results	P1	High	Medium
 Carbon Data Import API (top categories)	P1	Medium	High

P0 = Must-have for MVP launch. P1 = Fast-follow feature.

Our strategy prioritizes "Strategic Investments" and "Quick Wins" to ensure the MVP is both impactful and feasible.

“ Our MVP focuses on **signals over saturation** — giving users actionable insights now. ”

Post-MVP Features (6–18 Months)

Smart Alternatives Engine

Boosts eco-friendly product swaps and increases Average Order Value (AOV).

Personal Carbon Dashboard

Improves user retention by gamifying personal impact over time.

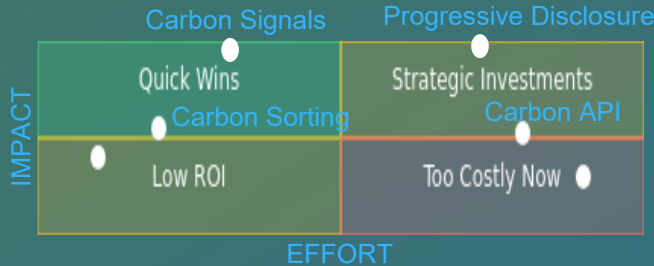
Supplier Portal

Scales data accuracy and automates verified carbon data collection.

Carbon-Optimized Shipping

Adds a critical sustainability lever at the final purchase stage.

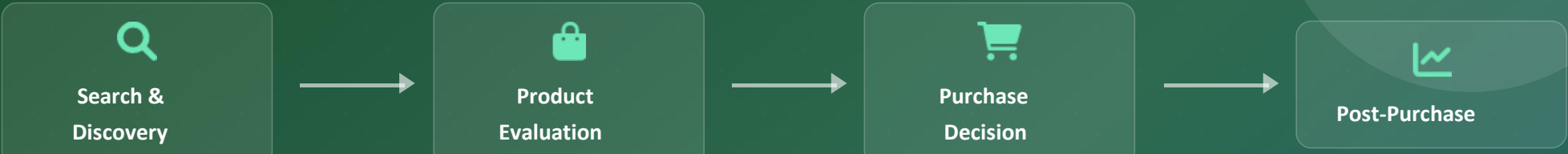
Feature Prioritization Strategy



Quick Wins Strategic Investments Low ROI Too Costly Now

Carbon-Conscious Journey

Seamless UX from Discovery to Impact



Search & Discovery

Carbon Filter

Carbon Signal

Maya: 5/5

Alex: 3/5

Carbon signals in listings help users identify sustainable options early in the journey.

Carbon Filters

Product Evaluation

Basic Rating

Detail Disclosure

Maya: 5/5

Alex: 4/5

Progressive disclosure allows users to see carbon data without overwhelming detail.

Progressive Disclosure

Purchase Decision

Smart Alternatives

Offset Option

Maya: 3/5

Alex: 5/5

Smart alternatives engine suggests eco-friendly swaps based on user preferences.

Smart Alternatives

Post-Purchase

Order Impact

Dashboard

Maya: 5/5

Alex: 3/5

Carbon dashboard shows personal impact and encourages continued sustainable choices.

Personal Dashboard

Key Touchpoints

Search & Discovery

Carbon filters and signals in listings

Product Evaluation

Progressive disclosure and comparison tools

Purchase Decision

Smart alternatives and offset options

Post-Purchase

Order-level impact and dashboard

"Our solution empowers users to make sustainable choices without friction — transforming regulatory compliance into a competitive advantage."

From Signals to Action: UX for Seamless Carbon Awareness

Our design follows the principle of **layered simplicity**: quick cues for most users, deep data for those who seek it.

Search Result Card

A+ CARBON RATING

Compact carbon signal

Tooltip on hover reveals: "35% below average"

Green Sneakers – Brand X

★★★★☆ (112)

₹3,499

- ✓ Color-coded badge provides immediate visual impact
- ✓ Tooltip on hover offers deeper context without requiring a click
- ✓ Integrates carbon as a standard product attribute

Product Detail Page

Progressive disclosure

A+ Carbon Footprint

3.6kg CO₂

35% below average

Product Features

Smart Alternatives

View Breakdown

Add to Cart

Offset CO₂

- ✓ Primary visual signal above the fold
- ✓ "View Breakdown" link invites deeper exploration
- ✓ Smart alternatives module addresses comparison needs

Carbon Breakdown Modal

Carbon Breakdown

Third-party verified

Stage	Emissions	%
Materials	2.1 kg	42%
Production	0.9 kg	18%
Shipping	0.4 kg	8%
Usage	0.2 kg	4%
Disposal	0.0 kg	0%

See Methodology

Deep-dive data for power users

Total: 3.6 kg CO₂

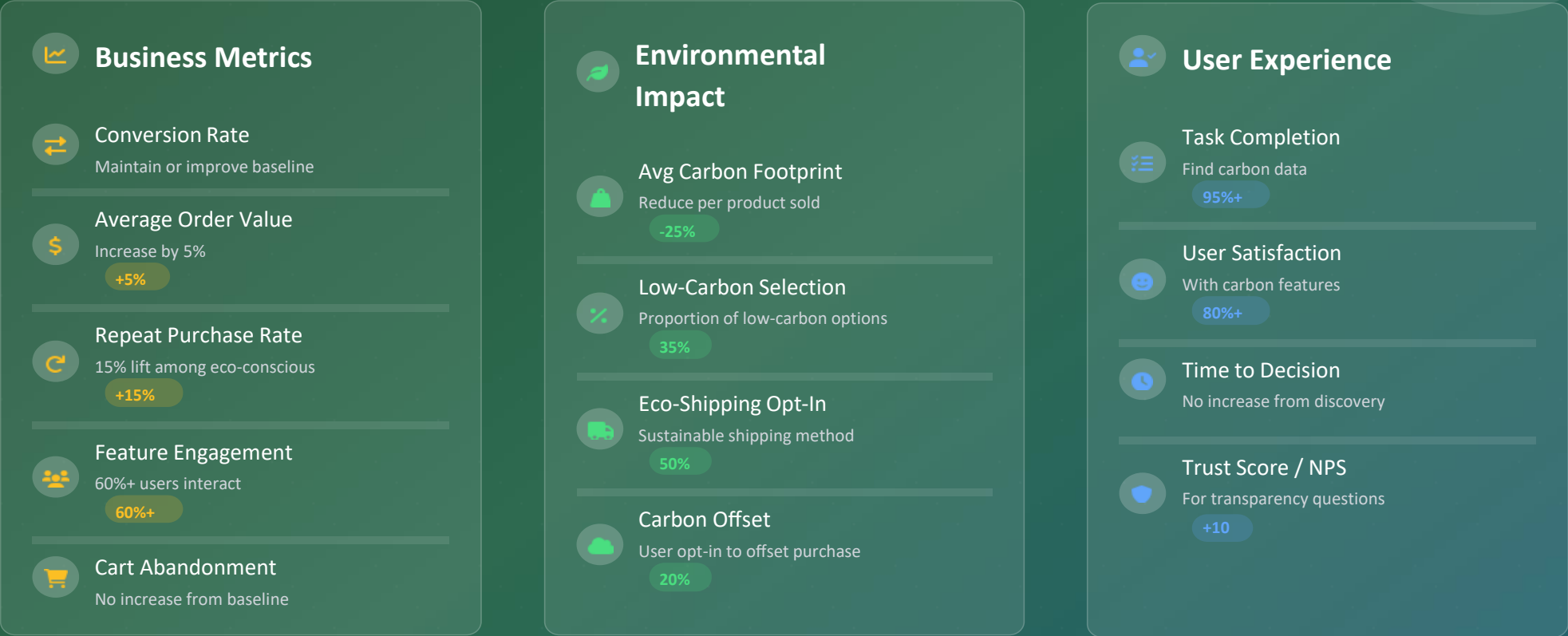
- ✓ Accessible via "View Breakdown" link on PDP
- ✓ Third-party verification badge builds trust
- ✓ Methodology link provides transparency

“

When transparency is elegant, users engage without effort.

”

Measuring Impact: Business, Sustainability, and User Outcomes



Success is measured across three pillars: business viability, environmental impact, and user satisfaction

From MVP to Movement: The Road to Sustainable E-Commerce

🗺️ 18-Month Product Roadmap

1

Foundation (0-6 Months)

Establishing core signals and backend infrastructure with minimal disruption.

✔️ Carbon Signal Indicators for top categories

✔️ Basic Progressive Disclosure framework

✔️ Initial Data Platform with top 20 suppliers

🎯 Goal: Establish user trust and achieve quick adoption

2

Expansion (6-12 Months)

Scaling solution across platform and introducing intelligent features.

✔️ Platform-wide carbon signal indicators

✔️ Smart Alternatives Engine (v1)

✔️ Sustainable Filtering & Sorting

✔️ Basic Personal Carbon Dashboard

🎯 Goal: Integrate sustainability into core user journey

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Optimization (12-18 Months)

Deepening engagement with sophisticated tools and personalization.

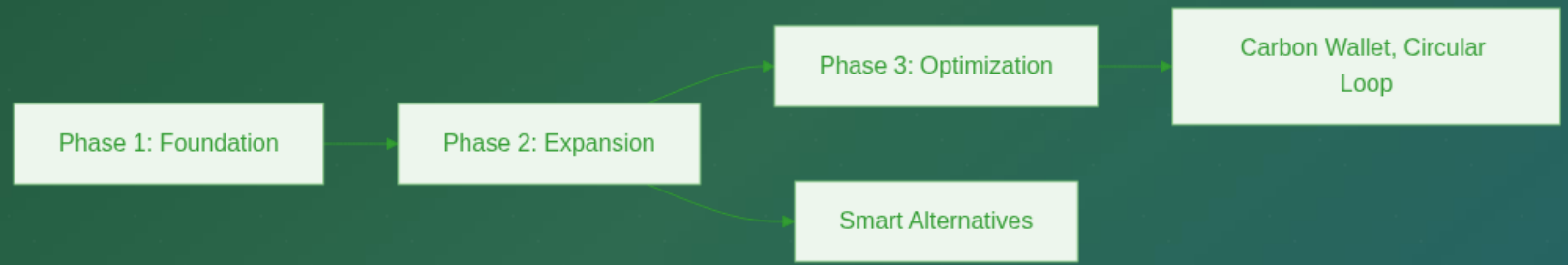
✔️ Full Progressive Disclosure Framework

✔️ Cross-Category Comparison

✔️ Carbon-Optimized Shipping

✔️ Enhanced User Profiles with carbon history

🎯 Goal: Build sustainable shopping habits and increase retention



🏡 5-Year Vision (2035-Ready Platform)

👤

Personal Carbon Budgeting

Dynamic "carbon wallet" tracking consumption against personalized goals. AI provides proactive recommendations for staying within budget.

🏭

Supplier Certification Ecosystem

Open, blockchain-verified system for suppliers to submit and update product-level carbon data, creating a trusted transparent pipeline.

🔄

Carbon Credits Marketplace

Seamless purchase of verified carbon removal credits at checkout, empowering users to offset residual emissions and take action.

♻️

Closed-Loop Commerce

Integration of circular economy principles with automated prompts for product repair, resale, and recycling, shifting from linear to circular model.

“ Our vision is not just compliance, but redefining commerce for a climate-conscious future. ”

De-Risking the Vision: What Could Go Wrong and What We'll Do About It

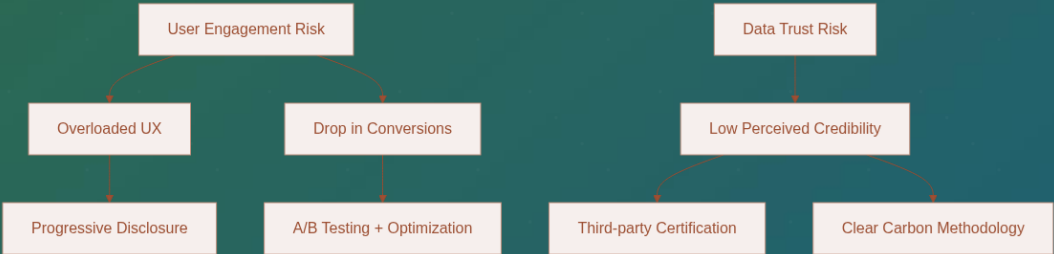
Key Risks & Mitigation

Risk	Mitigation Strategy
 Inaccurate Data High impact, Medium probability	 Implement hybrid data model with third-party verification and clear labeling
 Low User Adoption High impact, Medium probability	 Use progressive disclosure UX to reduce cognitive load
 Negative Impact on Conversion Medium impact, Medium probability	 A/B test all UX changes to ensure carbon info enhances purchasing decisions
 Supplier Reluctance	 Develop supplier onboarding toolkit with incentives for

Core Assumptions & Validation

"We've identified and planned for the most probable challenges, ensuring this solution is not only visionary but also resilient, scalable, and realistic."

Assumption	Validation Plan
 Users desire carbon data if accessible	 Track feature engagement and click-through rates
 Carbon signals won't harm conversion	 Run controlled A/B tests on product pages
 Users act on alternative recommendations	 Track selection rate and impact on AOV



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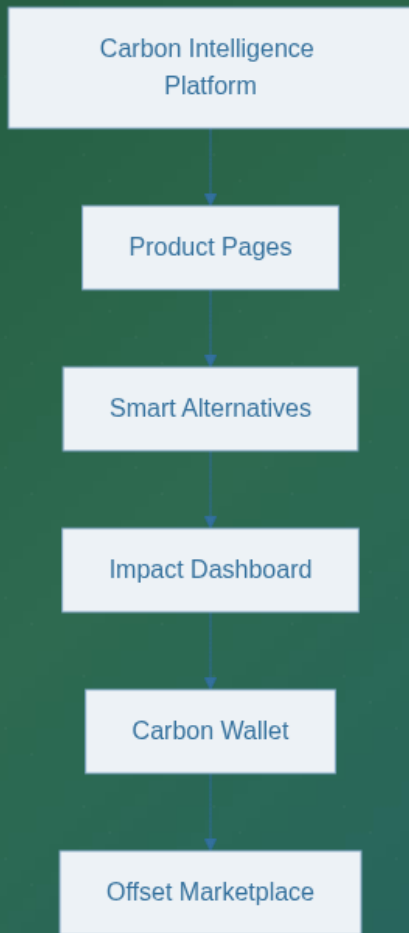
We're not just adding carbon labels. We're redesigning commerce for a climate-conscious future.

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Making Carbon Clarity the New Standard in E-Commerce

Key Takeaways

- **Empowers Users, Protects UX:** Clear carbon signals for all users with deep data for the curious without cognitive overload.
- **Turns Regulation into Brand Value:** Early action establishes market leadership and builds trust with eco-conscious consumers.
- **Balances Intention with Action:** Progressive disclosure, smart alternatives, and seamless nudges address the "intention-action gap."
- **Ready to Scale:** Phased roadmap begins with high-impact MVP and evolves into a future-proof platform for 2035 and beyond.



Next Steps

- **User Testing & Partner Pilots (Q3 2025)**
Begin qualitative testing with primary personas and launch a pilot program with key suppliers.
- **KPI Benchmarking & Iteration**
Establish baseline metrics for conversion, AOV, and engagement to measure MVP impact.
- **Supplier Onboarding Framework**
Develop tools and documentation needed to scale data collection from supply chain partners.
- **Long-Term Vision Refinement**
Continue R&D for future-state features including personal carbon wallets, impact badges, and circular commerce.



Carbon Clarity: Where Transparency Meets Action